THE JOURNAL OF BIOLOGICAL CHEMISTRY
Vol. 255, No. 9, Issue of May 10, pp. 3931-3934, 1980
Printed in U.S.A.

Kinetics of Association of Serine Proteinases with Native and Oxidized α-1-Proteinase Inhibitor and α-1-Antichymotrypsin*

(Received for publication, May 29, 1979, and in revised form, October 16, 1979)

Keith Beatty,‡ Joseph Bieth,§ and James Travis‡

From the ‡ Department of Biochemistry, University of Georgia, Athens, Georgia 30602 and the § Faculté de Pharmacie, Laboratoire d'Enzymologie, Université Louis Pasteur, Strasbourg, 67000 France

The association rate constants for the interaction of α-1-proteinase inhibitor, oxidized α-1-proteinase inhibitor, and α-1-antichymotrypsin with several mammalian serine proteinases have been determined. The results indicate that leukocyte elastase reacts more rapidly with α-1-proteinase inhibitor than any other proteinase tested, while leukocyte cathepsin G shows the strongest association with α-1-antichymotrypsin. Oxidation of the critical methionine residue of α-1-proteinase inhibitor reduces the association with leukocyte elastase by a factor of more than 2000 and also lowers the association with all of the other enzymes tested with the exception of chymotrypsin. Significantly, oxidation completely abolishes any interaction of α-1-proteinase inhibitor with porcine elastase, human plasmin, or human thrombin. These data support previous results (Johnson, D., and Travis, J. (1979) *J. Biol. Chem.* 254, 4022-4026) which indicated that oxidation of human α-1-proteinase inhibitor *in vivo* could reduce the effectiveness of this inhibitor in controlling proteolysis. In the lung, in particular, oxidizing agents of both chemical and biological sources could, indirectly, augment elastolysis in this tissue, resulting in the development of pulmonary emphysema.

Normal human plasma contains at least seven proteinase inhibitors which account for more than 10% of the total protein in this tissue (1). Two of these inhibitors, α-1-proteinase inhibitor (α-1-PI)[1] and α-1-antichymotrypsin (α-1-Achy), have been isolated to homogeneity in this laboratory (2, 3) and the former shown to contain a methionyl residue at its reactive center (4). Furthermore, oxidation of this critical amino acid has been found to either inactivate or delay the inhibitory activity of the inhibitor toward several serine proteinases (5). While the specific function of either inhibitor *in vivo* is not known, it is believed that α-1-PI may represent an important part of the defense mechanism of the lung toward proteolytic attack (6).

Both α-1-PI and α-1-Achy inhibit proteinases in a similar manner, with inhibition being strictly proportional to inhibitor concentration, suggesting that these proteins act as pseudoirreversible inhibitors (2, 7). Literature is lacking information concerning K_I, presumably because reversibility of the inhibitor·enzyme complex, a prerequisite for determining K_I, has never been achieved. This is either due to a very low dissociation constant (K_I 10^{-10} M) or to a stabilization of the complex by covalent bond or tetrahedral complex formation between enzyme and inhibitor. In fact, the presence of either type of interaction between the proteinase active center serine residue and a carbonyl group at the active site of the inhibitor has already been suggested to account for the extreme stability of α-1-PI·enzyme complexes in the presence of denaturing agents (8, 9).

The usual way of defining the specificity of a protein proteinase inhibitor is to determine its K_I for a number of proteinases. Since this is not possible with either α-1-PI or α-1-Achy, we have decided to investigate the specificity of each by measuring the rate constants of association with different proteinases. The present investigation was, however, also aimed at gaining a better insight into the physiological function of both inhibitors. It is commonly believed that if a protein inhibits a proteinase *in vitro*, it does also *in vivo*. This condition is, of course, necessary but not sufficient. An important factor is the delay time of inhibition *in vivo*. If this is too long, the proteinase may have enough time to reach its substrate, thus rendering the inhibitor inefficient. Alternatively, the enzyme could inactivate the inhibitor by proteolytic attack at a site(s) remote from the active site.

The inhibitors utilized in this study have no proteolytic activity or esterase activity and do not interfere with the assay by increasing proteinase activity. They do inhibit the proteolytic and esterolytic activity of free enzyme by forming an inhibitory complex between enzyme and inhibitor in a 1:1 molar ratio (2, 7). This paper reports the rate of formation of this complex in a large number of enzyme-inhibitor combinations. A significant finding is that the rate of complex formation varies more than a million-fold between the slowest and the most rapid combinations. A corollary to this finding is that the mechanism of inhibition is independent of the specificity of the enzyme.

EXPERIMENTAL PROCEDURES

Materials

The human proteins α-1-PI (2), α-1-Achy (3), leukocyte cathepsin G (4), leukocyte elastase (10), cationic and anionic trypsin (11, 12), chymotrypsin I (13), and thrombin (14), as well as shrimp trypsin (15), were prepared by techniques developed in this laboratory. Rabbit trypsin was a gift of Dr. David Johnson (16), East Tennessee State University. Bovine trypsin and chymotrypsin were obtained from Worthington and porcine trypsin from Sigma. Human plasminogen was isolated by the method of Deutsch and Mertz (17). After activation by streptokinase, plasmin was purified by affinity chromatogra-

* This research was supported in part by grants from the National Institutes of Health and The Council for Tobacco Research to J. T. and by a grant from Institut National de la Santé et de la Recherche Médicale to J. B. The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "*advertisement*" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

[1] The abbreviations used are: α-1-PI, α-1-proteinase inhibitor; α-1-Achy, α-1-antichymotrypsin; MeO-Suc-Ala-Ala-Pro-Val-*p*-nitroanilide, methoxysuccinyl-L-alanyl-L-alanyl-prolyl-L-valyl-*p*-nitroanilide; Suc-Ala-Ala-Pro-Phe-*p*-nitroanilide, succinyl-L-alanyl-L-alanyl-prolyl-L-phenylalanyl-*p*-nitroanilide.

This is an Open Access article under the CC BY license.

TABLE I

Association rate constants for the interaction of α-1-PI, oxidized α-1-PI, and α-1-Achy with various proteinases

The constants were measured at 25°C, pH 8.0. Error was determined by least squares linear regression.

Enzyme (final molarity[a])	Substrate (final molarity)	k_{assoc}		
		α-1-PI	oxidized α-1-PI	α-1-Achy
		$M^{-1} s^{-1}$		
Human leukocyte elastase	MeO-Suc-Ala-Ala-Pro-Val-*p*-nitroanilide (0.3 mM)	$6.5 \pm 4.0 \times 10^7$	$3.1 \pm 0.2 \times 10^4$	
Bovine chymotrypsin (10 nM)	Suc-Ala-Ala-Pro-Phe-*p*-nitroanilide (0.2 mM)	$5.9 \pm 0.6 \times 10^6$	$2.4 \pm 0.2 \times 10^4$	$6.0 \pm 1.0 \times 10^4$
Human chymotrypsin 1 (14 nM)	Suc-Ala-Ala-Pro-Phe-*p*-nitroanilide (0.2 mM)	$5.4 \pm 0.3 \times 10^6$	$1.0 \pm 0.4 \times 10^6$	$1.0 \pm 0.1 \times 10^4$
Human cathepsin G (100 nM)	Suc-Ala-Ala-Pro-Phe-*p*-nitroanilide (4.0 mM)	$4.1 \pm 0.6 \times 10^5$	$6.4 \pm 0.3 \times 10^2$	$5.1 \pm 0.7 \times 10^7$
Bovine trypsin (100 nM)	Bz-L-Arg-OEt (5.0 mM)	$1.3 \pm 0.3 \times 10^7$	$8.2 \pm 0.4 \times 10^3$	
Rabbit trypsin (30 nM)	Bz-L-Arg-OEt (5.0 mM)	$1.0 \pm 0.1 \times 10^7$	$1.3 \pm 0.1 \times 10^4$	
Porcine elastase (14 nM)	Suc-Ala-Ala-Ala-*p*-nitroanilide (1.2 mM)	$1.0 \pm 0.2 \times 10^5$	0	
Human anionic trypsin (40 nM)	Bz-L-Arg-OEt (5.0 mM)	$7.3 \pm 1.8 \times 10^4$	$3.2 \pm 0.1 \times 10^4$	
Shrimp trypsin (30 nM)	Bz-L-Arg-OEt (5.0 mM)	$5.9 \pm 0.3 \times 10^4$	$7.6 \pm 0.5 \times 10^2$	
Porcine trypsin (120 nM)	Bz-L-Arg-OEt (5.0 mM)	$4.2 \pm 0.3 \times 10^4$	$5.0 \pm 0.2 \times 10^3$	
Human cationic trypsin (30 nM)	Bz-L-Arg-OEt (5.0 mM)	$1.1 \pm 0.2 \times 10^4$	$3.0 \pm 1.1 \times 10^3$	
Human plasmin	Bz-Phe-Val-Arg-*p*-nitroanilide (0.2 mM)	$1.9 \pm 0.1 \times 10^2$	0	
Human thrombin (3.4 μM)	Bz-L-Arg-OEt (5.0 mM)	$4.8 \pm 0.5 \times 10^1$	0	

[a] Indicated only when equimolar concentrations of enzyme and inhibitor were reacted.

phy.[2] Oxidized α-1-PI was prepared by *N*-chlorosuccinimide treatment of the native protein (18).

MeO-Suc-Ala-Ala-Pro-Val-*p*-nitroanilide and Suc-Ala-Ala-Pro-Phe-*p*-nitroanilide were gifts of Dr. James Powers, Georgia Institute of Technology, Atlanta, Ga. Bz-L-Arg-OEt was from Sigma and Bz-Phe-Val-Arg-*p*-nitroanilide from A. B. Bofors. Suc-$(Ala)_3$-*p*-nitroanilide was purchased from Bachem. The individual substrates used in the specific assay of each enzyme tested are given in Table I, together with their final concentrations in the assay medium.

Stock solutions of substrates were made in dimethylformamide, except for Suc-Ala-Ala-Pro-Phe-*p*-nitroanilide and MeO-Suc-Ala-Ala-Pro-Val-*p*-nitroanilide which were dissolved in dimethyl sulfoxide. Enzymes were kept in the following solutions prior to use: 0.001 N HCl (all pancreatic proteinases except elastase); 0.05 M Tris-HCl, pH 8.0 (porcine pancreatic elastase); 0.05 M sodium acetate, pH 5.0, and 0.5 M NaCl (leukocyte elastase and leukocyte cathepsin G); 0.25 M sodium phosphate, pH 6.5 (thrombin); and 0.02 M lysine, 0.15 M NaCl, and 20% glycerol, pH 3.0 (plasmin). α-1-PI, oxidized α-1-PI, and α-1-Achy were kept at pH 8.0 in 0.05 M Tris-HCl, 0.05 M NaCl.

Methods

Inhibitor Assays—Active site titration of porcine pancreatic trypsin was performed by the method of Chase and Shaw (19). This trypsin was then used to titrate α-1-PI solutions as follows. Trypsin (10 nmol) was reacted during 4 min at 25°C with the inhibitor (2 to 14 nmol) in 0.05 M Tris-HCl, 0.05 M NaCl, pH 8.0, after which free trypsin activity was measured with Bz-L-Arg-OEt (20). The inhibitor was then used as a secondary standard for determining the activity of all of the other proteinases used in this study. The enzyme and inhibitor concentrations were chosen high enough to ensure complete association within a few minutes (*i.e.* about 100 times higher than the concentrations indicated in Table I). Residual enzyme activities were then measured with the appropriate substrate. For example, the titration of human leukocyte elastase was performed as follows. The enzyme (0.7 nmol) was reacted for 1 min with variable amounts of α-1-PI in 0.05 M Tris-HCl and 0.5 M NaCl, pH 8.0. The residual enzyme activity was then measured by adding 20 μl of 0.060 M Suc-$(Ala)_3$-*p*-nitroanilide to 1.0 ml of reaction medium (21). With this enzyme, as well as with all others tested, straight inhibition curves were obtained which allowed the concentration of active sites in each proteinase to be determined precisely.

The active site titration of bovine α-chymotrypsin was also measured, as described by Kezdy and Kaiser (22), except that *t*-Boc-L-Tyr-*p*-nitroanilide was used as substrate. This enzyme was used directly to calculate the activity of α-1-Achy, following the procedure described above for the titration of α-1-PI.

Except for human plasmin and leukocyte elastase, the association rate constants were determined under second order conditions by allowing equimolar concentrations of enzyme and inhibitor to react for various periods of time before addition of substrate which stops the reaction and measures residual enzyme activity. In a typical experiment, the association reaction was started by mixing 20 μl of stock solution of enzyme into 980 μl of inhibitor solution contained in a semimicrocuvette. After a given period of time, the reaction was stopped by addition of 20 μl of substrate solution. The final enzyme and substrate concentrations are indicated in Table I. The rate constant for the association of plasmin with α-1-PI was measured under pseudo-first order conditions by incubating plasmin (1.1 nmol) with a 20-fold molar excess of inhibitor in 0.05 M Tris-HCl, 0.05 M NaCl, pH 8.0, 25°C. After given periods of time, 200-μl aliquots of the mixture were diluted into 800 μl of substrate contained in a spectrophotometer cuvette and residual activity measured. The rate constant for human leukocyte elastase was derived from competition experiments with bovine α-chymotrypsin. α-1-PI (70 nmol) was incubated with mixtures of elastase (70 nmol) and varying quantities of chymotrypsin (5 to 100 nmol) in a total volume of 1.0 ml. After 1 min, the elastase activity was measured by addition of 20 μl of MeO-Suc-Ala-Ala-Pro-Val-*p*-nitroanilide (15 mM) dissolved in dimethylsulfoxide.

In deriving association rate constants, the random error of intercepts was calculated by a least squares linear regression program on a NOVA II computer. The propagation of error in the computed result was determined by accepted rules. The random error was found to vary from a low of ±3% to a high of ±60%. It is reasonable to expect systematic errors of similar magnitudes. All values are probably accurate to within an order of magnitude and considering the large range of values which span seven orders of magnitude, this precision is sufficient to allow valid comparisons between values.

RESULTS AND DISCUSSION

It has been claimed that the inhibition of pancreatic proteinases by α-1-PI occurs too rapidly to measure experimentally (23). This is because too high concentrations of enzyme and inhibitor are normally used in such experiments. With our experimental conditions, however, time-dependent inhibition was observed for all proteinases except leukocyte elastase. To analyze our data, we assume that the proteinases (E) react with the inhibitors α-1-PI, oxidized α-1-PI, and α-1-Achy (I) according to Equation 1:

$$E + I \underset{k_{dissoc}}{\overset{k_{assoc}}{\rightleftharpoons}} EI \qquad (1)$$

The rate of association is given by Equation 2:

$$-\frac{d(E)}{dt} = k_{assoc}(E)(I) - k_{dissoc}(EI) \qquad (2)$$

The second term of this equation may be neglected since the back reaction could never be observed experimentally and since in our hands the inhibition obeyed second order kinetics

[2] K. Beatty and J. Travis, unpublished results.

up to at least 90% completion of the reaction.[3] For those data obtained by reacting equimolar concentrations of enzyme and inhibitor, the simplified form of Equation 2 integrates to:

$$\frac{1}{(E)} = k_{assoc} \cdot t + \frac{1}{(E^0)} \quad (3)$$

where (E) and (E^0) stand for free and total enzyme concentrations, respectively. For the sake of convenience, however, we plotted enzyme rates (v) instead of concentrations. Plots of v^{-1} *versus* t (which were linear in all cases) enabled us to determine $t_{0.5}$, the half-life time of reaction which is related to k_{assoc} through the following:

$$t_{0.5} = \frac{1}{k_{assoc}(E^0)} \quad (4)$$

For plasmin, which is an unstable enzyme and which, in addition, reacts slowly with α-1-P1, we used a 20-fold molar excess of inhibitor over enzyme in order to increase the inhibition rate. Under these conditions, the reaction may be considered as pseudo-first order with respect to enzyme. Indeed, a semilogarithmic plot of v *versus* t gave a straight line from which $t_{0.5}$ and, hence, k_{assoc} could be determined (Equation 5):

$$t_{0.5} = \frac{0.693}{k_{assoc}(I^0)} \quad (5)$$

The case of human leukocyte elastase was more complicated. Even with the very sensitive new substrate MeO-Suc-Ala-Ala-Pro-Val-*p*-nitroanilide (25), the enzyme concentration could not be made low enough to obtain time-dependent inhibition. k_{assoc} was, therefore, measured indirectly by allowing elastase and bovine chymotrypsin to compete for the binding with either α-1-PI or oxidized α-1-PI. We measured the elastase activity of mixtures formed of constant amounts of elastase and inhibitor and variable amounts of chymotrypsin. The reaction time (1 min) was long enough to enable the inhibitor to react with either one of its partners. Under these conditions, the data could be analyzed according to Equation 6 (26):

$$\frac{k_{assoc}}{k'_{assoc}} = \frac{\log (a/(a-x))}{\log (b/(b-y))} \quad (6)$$

where k_{assoc} and k'_{assoc} are the second order rate constants for the association of inhibitor with elastase and chymotrypsin, respectively; a and b are the total concentrations of elastase and chymotrypsin; and x and y are the concentrations of inhibitor bound to elastase and chymotrypsin, respectively. The values of x were determined from the elastase activities of the mixtures and the values of y were calculated (total inhibitor concentration $- x$). A plot in accordance with Equation 6 gave a straight line whose slope yielded k_{assoc}/k'_{assoc}. Due to the active site titration methods, which all refer to a primary standard (porcine trypsin), it may be concluded that all of the k_{assoc} data possess a high degree of internal consistency.

As can be seen from Table I, k_{assoc} increases by more than six orders of magnitude between thrombin and leukocyte elastase when they interact with α-1-PI. The k_{assoc} value for this latter proteinase is the highest ever reported for a proteinase inhibitor system (*e.g.* compared with 8.2×10^6 M^{-1} s^{-1} for the association of trypsin with soybean inhibitor (27)) and is only about 15 times lower than the diffusion-controlled limit. Thus, α-1-PI, which was formerly named α-1-antitrypsin (28), and later thought to be a more specific chymotrypsin inhibitor (29), must now be considered as a specific human leukocyte elastase inhibitor since it reacts 10 and 1000 times faster than with human chymotrypsin and human trypsin, respectively. Furthermore, α-1-Achy, which shows a similar k_{assoc} with cathepsin G, should probably be recognized as the specific inhibitor of this enzyme. In fact, both sets of results are in complete agreement with competition experiments in which α-1-PI, α-1-Achy, chymotrypsin, and cathepsin G were mixed together and shown to specifically interact to form α-1-PI·chymotrypsin complexes and α-1-Achy·cathepsin G complexes (7). The data obtained with bovine and human trypsins and chymotrypsins are also in good agreement with those found previously with crude α-1-PI (30).

The oxidation of human α-1-PI had a profound effect on its ability to interact with most serine proteinases, chymotrypsin being the major exception. In particular the k_{assoc} with elastolytic enzymes was drastically altered, being reduced to zero for the pancreatic enzyme and more than 2000-fold less for leukocyte elastase. Since α-1-PI deficiency frequently leads to emphysema (6) and leukocyte elastase can induce lung damage in animals (31), any alteration in the function of this inhibitor in individuals with normal plasma protein levels could lead, ultimately, to the development of lung disease. In particular, as already has been shown in this laboratory (18, 32), both chemical and biological oxidants can readily convert native α-1-PI to the oxidized form, thereby reducing the effective interaction of this inhibitor with leukocyte elastase. Should this occur *in vivo*, degradation of lung elastin might readily occur.

The conclusion that the physiological function of α-1-PI is to protect lung tissue against the deleterious effects of elastase liberated from leukocytes during phagocytosis is valid only if the inhibition is fast enough *in vivo* to prevent any competition between α-1-PI and lung elastin. From the values of k_{assoc} and of the physiological concentration of α-1-PI in plasma (2.5×10^{-5} M), a pseudo-first order rate constant (*i.e.* $k_{assoc}(I^0)$) of 1.6×10^3 s^{-1} may be calculated. This shows that leukocyte elastase is inhibited *in vivo* with a half-life of only 420 μs and confirms by quantitative means that, indeed, α-1-PI must play an important physiological role in the prevention of emphysema. A k_{assoc} value of 10^4 M^{-1} s^{-1} or less would have ruled out such a function.

If we apply an identical reasoning to the slowly reacting enzymes, plasmin and thrombin, we find that their half-life times in plasma are 2.4 min and 9.6 min, respectively. Inspection of such high numbers raises the question: can α-1-PI be considered as a physiological inhibitor of thrombin and plasmin? Plasmin is generated from plasminogen at the end of the coagulation process in order to digest the fibrin clot. The time of 2.4 min is probably high enough to enable plasmin to reach its substrate and low enough to enable α-1-PI to destroy plasmin activity within a reasonable period of time after it has accomplished its fibrin-degrading function. α-1-PI may, there-

[3] We are aware of the fact that Scheme 1 might be an oversimplification. The reaction of E and I might proceed by the formation of a loose EI^* complex with transforms into a stable EI complex as shown in the soybean trypsin inhibitor system by Luthy *et al.* (24). The microscopic constants describing this reaction are given in the following scheme:

$$E + I \underset{k^{-1}}{\overset{k_1}{\rightleftharpoons}} EI^* \xrightarrow{k_2} EI$$

Hence, k_{assoc} defined in our Scheme 1 is related to these constants in the following relationship:

$$k_{assoc} = \frac{k_2 k_1}{k_{-1}}$$

However, as we are unable to detect, kinetically, the EI^* complex and the kinetics was always second order, we feel justified in analyzing our results according to Scheme 1.

fore, play a physiological role in regulating plasmin activity, together with other plasmin inhibitors. Thrombin is generated from prothrombin in order to transform fibrinogen into insoluble fibrin. Its action must be strictly limited in time in order to avoid generalized fibrinogen consumption. Its activity must, therefore, be rapidly destroyed. From these considerations it may be inferred that α-1-PI does not play any physiological role in the regulation of thrombin activity.

α-1-PI has also been thought to protect plasma proteins against the deleterious effect of trypsin which is massively liberated into the blood stream in the course of acute pancreatitis. From our data, we find that the half-life time of cationic trypsin is about 3 s. During this long delay time of inhibition, the enzyme may activate coagulation and hypotensive peptide-generating proenzymes whose effect will lead to the so-called "pancreatic shock." Our previous statement that cationic trypsin must be considered as a major mediator of pancreatic shock (33) is, thus, entirely confirmed by the data of the present studies. Anionic trypsin is inactivated with a half-life of about 0.4 s. α-1-PI is, thus, a more probable physiological inhibitor of this proteinase. It must, however, be recalled that this enzyme represents only about 20% of the total pancreatic trypsin.

Inspection of Table I reveals that the decreasing order of reactivity of proteinases with α-1-PI is: human leukocyte elastase > chymotrypsins (including cathepsin G which has chymotrypsin specificity (34)) > trypsins (including thrombin and plasmin) = pancreatic elastase. These results, however, cannot be correlated with the active center sequence of α-1-PI which has recently been shown to have the P_4 to P_1 sequence ... Ala-Ile-Pro-Met ... (4), since the substrate specificities of each group of enzymes are not, generally, directed toward this amino acid. However, it is interesting to note that the synthetic substrate MeO-Suc-Ala-Ile-Pro-Met-*p*-nitroanilide, which mimics the active center of α-1-PI, is rapidly cleaved by human leukocyte elastase ($k_{cat}/K_m = 4000\ M^{-1}\ s^{-1}$), bovine chymotrypsin ($k_{cat}/K_m = 4400\ M^{-1}\ s^{-1}$), and human cathepsin G ($k_{cat}/K_m = 710\ M^{-1}\ s^{-1}$) (24) but not by trypsin, which cleaves preferentially substrates with P_1 Arg or Lys residues. We have, thus, a rationale for the lower k_{assoc} values of the trypsins with respect to that of the other proteinases. It must, however, be added that factors other than the primary recognition site of α-1-PI (*i.e.* the P_4 to P_1 sequence) play an important role in the association of this protein with proteinases. Presumably, this same argument also holds true for α-1-Achy. However, the limited specificity of this inhibitor may be an advantage in determining differences in the mechanism of action of plasma proteinase inhibitors *versus* those from other plant and animal sources.

Acknowledgments—We wish to thank Dr. David Johnson for the gift of rabbit trypsin and Dr. James Powers for substrates.

REFERENCES

1. Heimburger, N., Haupt, H., and Schwick, H. (1971) *Proceedings of the First International Research Conference on Proteinase Inhibitors (1970)* pp. 1–21, Walter de Gruyter, New York
2. Pannell, R., Johnson, D., and Travis, J. (1974) *Biochemistry* **13,** 5439–5445
3. Travis, J., Garner, D., and Bowen, J. (1978) *Biochemistry* **17,** 5647–5651
4. Johnson, D., and Travis, J. (1978) *J. Biol. Chem.* **253,** 7142–7144
5. Johnson, D., and Travis, J. (1979) *J. Biol. Chem.* **254,** 4022–4026
6. Laurell, C. B., and Eriksson, S. (1963) *Scand. J. Clin. Invest.* **15,** 132–140
7. Travis, J., Bowen, J., and Baugh, R. (1978) *Biochemistry* **17,** 5651–5656
8. Moroi, M., and Yamasaki, M. (1974) *Biochim. Biophys. Acta* **359,** 130–141
9. Cohen, A. B., Gruenke, L. D., Craig, J. C., and Geczy, D. (1977) *Proc. Natl. Acad. Sci. U. S. A.* **74,** 4311–4314
10. Baugh, R. J., and Travis, J. (1976) *Biochemistry* **15,** 836–841
11. Travis, J., and Roberts, R. C. (1969) *Biochemistry* **8,** 2884–2889
12. Mallory, P. A., and Travis, J. (1973) *Biochemistry* **12,** 2847–2851
13. Coan, M. H., and Travis, J. (1972) *Biochim. Biophys. Acta* **268,** 207–211
14. Matheson, N. R., and Travis, J. (1976) *Biochem. J.* **159,** 495–502
15. Gates, B. J., and Travis, J. (1969) *Biochemistry* **8,** 4483–4489
16. Johnson, D. (1976) *Biochim. Biophys. Acta* **424,** 482–487
17. Deutsch, D. G., and Mertz, E. T. (1970) *Science* **170,** 1095–1096
18. Shechter, Y., Burstein, Y., and Patchornik, A. (1975) *Biochemistry* **14,** 4497–4503
19. Chase, T., and Shaw, E. (1969) *Biochemistry* **8,** 2212–2223
20. Schwert, G. W., and Takenaka, Y. (1955) *Biochim. Biophys. Acta* **16,** 570–575
21. Bieth, J., Spiess, B., and Wermuth, G. (1974) *Biochem. Med.* **11,** 350–357
22. Kezdy, F. J., and Kaiser, E. T. (1970) *Methods Enzymol.* **19,** 3–20
23. Cohen, A. B. (1975) *Biochim. Biophys. Acta* **391,** 193–200
24. Luthy, J. A., Praissman, M., Finkenstadt, W. R., and Laskowshi, M., Jr. (1973) *J. Biol. Chem.* **248,** 1760–1771
25. Nakajima, K., Powers, J. C., Ashe, B. M., and Zimmerman, M. (1979) *J. Biol. Chem.* **254,** 4027–4032
26. Vincent, J.-P., and Lazdunski, M. (1972) *Biochemistry* **11,** 2967–2977
27. Laskowski, M., Jr., and Sealock, R. W. (1971) in *The Enzymes* (Boyer, P. D., ed), pp. 375–473, Academic Press, New York
28. Heimburger, N., and Haupt, H. (1966) *A. Gesamte Inn. Med. Grenzeed* **21,** 193–201
29. Rimon, A., Shamash, Y., and Shapiro, B. (1966) *J. Biol. Chem.* **241,** 5102–5107
30. Aubry, M., and Bieth, J. (1977) *Clin. Chim. Acta* **78,** 371–380
31. Janoff, A., Sloan, B., Weinbaum, G., Damiano, V., Sandhaus, R. A., Elias, J., and Kimbel, P. (1977) *Am. Rev. Respir. Dis.* **115,** 461–478
32. Matheson, N. R., Wong, P. S., and Travis, J. (1979) *Biochem. Biophys. Res. Commun.* **88,** 402–409
33. Aubry, M., Travis, J., and Bieth, J. (1977) *Biochem. Med.* **18,** 360–370
34. Blow, A. M. J., and Barrett, A. J. (1977) *Biochem. J.* **161,** 17–19